Analysis of Technical Performance of Commercially Available Rice Combine Harvesters (Case Analysis)

Introduction/Background:

The goal of modern farming system is to reduce farming cost and to economize energy consumption. Optimal design and matching of agricultural machines proportionate to the present power must be considered in order to achieve this goal. This leads to an increase in farm efficiency, time saving in the farm operation, maximizing the use of tractor power and promoting value for money.

This is consistent with the goal of the RCEF Mechanization Program that is to, a) Reduce the production cost of farmer-users by Php2-3 per kilogram using precise, effective and complete system of mechanized production technologies, and b) Reduce the postproduction losses of farmer-users by 3-5% using appropriate and efficient postproduction technologies.

The technical requirement of a rice combine harvester as needed for RCEF Mechanization Program were based on a) estimates and analysis; b) our actual experiences and testimonies of farmers; c) actual observations and research on the performance of different models of commercially available rice combine harvesters; d) feedbacks and reports from the DA implementers nationwide; and e) intended use and purpose of the technology.

Objectives:

1. Estimate theoretical fuel consumption and theoretical field capacity of Rice Combine harvesters
2. Estimate cost differences in terms of theoretical fuel consumption and theoretical field capacity and other parameters.

Methodology:

The following parameters were considered in the analysis of field capacity and fuel consumption. The most ideal comparison of field capacity and fuel consumption of a combine harvester are conducting a field test with the SAME soil condition, crop condition, skills of operator, size and terrain of field, other technical parameters and environmental conditions. With these, the true fuel consumption and capacity can be determined.

AMTEC test established the actual fuel capacity and actual field capacity of different combine harvesters based on the operator/requisitionaire setting. The test of different combine harvesters were conducted with DIFFERENT soil condition, crop condition, skill of operator, size and terrain of field, other technical parameters and environmental conditions. Hence, the test result for fuel consumption and field capacity were affected by other factors which are not solely inherent to the machine itself.

The analysis and estimates of fuel consumption and field capacity using the standard computation of theoretical field capacity and theoretical fuel consumption is one way of establishing the performance of a machine.

Results of Estimates and Analysis:

The theoretical field capacity and theoretical fuel consumption are presented below:

1. Theoretical field capacity

   The theoretical field capacity in ha/hr can be computed based on formula FC = SWE, where FC is the Field Capacity in ha/hr, W is the width of cut in m, S is the speed in kph and E is the engine efficiency at 80%.

   In computing the theoretical field capacity of combine harvesters with rated power ranges from 60hp – 88 hp, a given speed at 4kph, 5kph and 6kph were used and the width of cut based on the manufacturers technical specification as indicated in their respective brochure. Also, with the assumption that these combine harvesters will be tested in the same area, shape of the test site, condition of the soil, condition of the crop and the same level of skills of operators.

   Based on data gathered, the width of cut of combine harvesters available in Philippine market ranges from 1.975 m to 2.075 m. Considering an operating speed of 4kph, 5kph and 6 kph, the combine harvesters with power rating ranges from 60-88 hp have comparable theoretical field capacity in ha/hr. As indicated in the respective manufacturer's brochure, the combine harvester with rated power of 83 hp has the least width of cut of 1.975 m while the combine harvester with rated power of 68 hp has a cutting width of 2.075 m. The combine harvesters with rated power of 87hp and 88 hp have a width of cut of 2m.

Table 1. Theoretical Field Capacity at speed of 4kph

Brand*	Power Rating	Cutting Width, m	Operating Speed, kph	Theoretical Field Capacity, ha/hr		
				75 % eff	80% eff	90% eff
A	68	2.075	4	0.623	0.664	0.747
B	70	2	4	0.600	0.640	0.720
C	79	2.06	4	0.618	0.659	0.742
D	83	1.975	4	0.593	0.632	0.711
E	87	2	4	0.600	0.640	0.720
F	88	2	4	0.600	0.64	0.720

*Based on commercially available brands

Table 2. Theoretical Field Capacity at speed of 5kph

Brand*	Power Rating	Cutting Width, m	Operating Speed, kph	Theoretical Field Capacity, ha/hr		
				75 % eff	80% eff	90% eff
A	68	2.075	5	0.778	0.830	0.934
B	70	2	5	0.750	0.800	0.900
C	79	2.06	5	0.773	0.824	0.927
D	83	1.975	5	0.741	0.790	0.889
E	87	2	5	0.750	0.800	0.900
F	88	2	5	0.750	0.8	0.900

*Based on commercially available brands

Table 3. Theoretical Field Capacity at speed of 6kph

Brand*	Power Rating	Cutting Width, m	Operating Speed, kph	Theoretical Field Capacity, ha/hr		
				75 % eff	80% eff	90% eff
A	68	2.075	6	0.934	0.996	1.121
B	70	2	6	0.900	0.960	1.080
C	79	2.06	6	0.927	0.989	1.112
D	83	1.975	6	0.889	0.948	1.067
E	87	2	6	0.900	0.960	1.080
F	88	2	6	0.900	0.96	1.080

*Based on commercially available brands

2. Fuel consumption

Fuel consumption is measured by the amount of fuel used during a specific time period. The most common measure of the energy efficiency of a tractor is referred to here as "specific volumetric fuel consumption" (SVFC), which is given in units of gallons per horsepower-hour (gal/hp-h). SVFC for diesel engines typically ranges from 0.0476 to 0.1110 gal/hp-h. (Grisco, 2014).

Formula used for the computation of fuel consumption using a 350 g/kw-hr (maximum SVFC per AMTEC recommendation) specific volumetric fuel consumption is as follows:

Formula 1: SFC x Fuel Density x (Engine power x efficiency) = FC, li/hr

Using formula 1, the table below shows the comparison of the computed theoretical fuel consumption in li/hr, of different combine harvesters with power rating ranges from 60 hp to 88 hp. Theoretical fuel consumption increases as the rated power of the combine harvester increases. Comparing the theoretical fuel consumption of the combine harvesters with power rating of 60 hp, 68hp and 83 hp to a combine harvester with power rating of 88 hp, have significant difference of 7.03 li/hr, 5.02 li/hr and 1.25 li/hr, respectively, or saved fuel.

Table 4. Theoretical fuel consumption of combine harvesters at 60-88 hp rated power

Brand*	Power Rating	Theoretical Fuel Consumption, li/hr
A	60	15.06
B	68	17.07
C	83	20.84
D	70	17.57
E	79	19.83
F	70	17.57
G	87	21.84
H	88	22.09

*Based on commercially available brands

Considering a full machine utilization of about eight (8) hours operation per day (10:00am to 6:00pm) and a 30 days harvest period per cropping season, the significant difference of theoretical fuel consumption of 60hp, 68 hp and 83 hp can be translated to an estimated amount saved of Php 151,839.69/yr, Php 108,456.92/yr and Php 27,114.23/year.

Table 5. Theoretical Fuel Consumption in Php/year

Brand*	Power Rating	Theoretical Fuel Consumption, li/hr	Cost of Fuel, Php/li	FC, P/hr	time of operation, hr/day	no of days/year	FC, P/year
A	60	15.06	45.00	677.86	8	60	325,370.77
B	68	17.07	45.00	768.24	8	60	368,753.54
C	83	20.84	45.00	937.70	8	60	450,096.23
D	70	17.57	45.00	790.83	8	60	379,599.23
E	79	19.83	45.00	892.51	8	60	428,404.85
F	70	17.57	45.00	790.83	8	60	379,599.23
G	87	21.84	45.00	982.89	8	60	471,787.62
H	88	22.09	45.00	994.19	8	60	477,210.46

*Based on commercially available brands

Under the RCEF Mechanization Program, for the initial 2B fund for year 1 target, about 356 units of combine harvester will be purchased and distributed nationwide. Considering the additional 3B fund for year 1 of which SARO has already been released to PHilMech, more or less the same number of combine harvester will be purchased and distributed nationwide. Assuming a total of 600 units of combine harvester for year 1 target, saved fuel can be directly translated to an estimated amount of Php 91,103,815.38/year, Php 65,074,153.85/year, and Php 16,268,538.46/year, respectively.

Table 6. Saved amount for fuel with respect to the estimated annual allocation for combine harvester

	Php	No of units, yr1	Total Saved Amount for Fuel, Php
FC difference of 60 hp to 88 hp, Php	151,839.69	600	91,103,815.38
FC difference of 68 hp to 88 hp, Php	108,456.92	600	65,074,153.85
FC difference of 85 hp to 83 hp, Php	27,114.23	600	16,268,538.46

3. Impact of heavy weight equipment to the rice field/soil hardpan

The weight of machine used in the rice field directly and/or indirectly affect to the soil/hardpan. The dimension (width and length) of the combine harvester track pad carrying the machine weight has effect on the load distribution to the soil. The combine harvester brand G with rated power of 87 hp and track pad dimension of .5m x 1.15 m has the highest draft force of 5443.48 kg/square meter while brand B has the least draft of 3529.41 kg/square meter.

Table 7. Draft in kg/square meter of commercially available combine harvesters

Brand Name	Power Rating, hp	Machine Weight	Track Pad			Draft, kg/square meter
			Width, m	Length,m	Area, m2	
A	60	2480	0.4	1.545	0.618	4012.94
B	68	3000	0.5	1.7	0.850	3529.41
C	83	3333	0.55	1.75	0.963	3462.86
D	70	3160	0.5	1.7	0.850	3717.65
E	79	3180	0.5	1.7	0.850	3741.18
F	70	3313	0.5	1.7	0.850	3897.65
G	87	3130	0.5	1.15	0.575	5443.48
H	88	2900	0.45	1.7	0.765	3790.85

4. Repair and Maintenance Cost

Repair cost are the expenditures for parts and labor for (1) installing replacement parts after a part failure and (2) reconditioning renewable parts as a result of wear. An Repair and Maintenance of %P/100 hr give numerical value that are easy to use for estimating machine cost. The repair and maintenance costs, percentage of list price (Hunt, 2016 - Table 4.5) for combine harvester (self-propelled) R&M is 1.33%.

Table 8 shows the comparison of repair and maintenance cost of a 60hp and 68 hp combine harvester with respected to the estimated machine cost (based on RFOs procurement and market research). Results show that higher machine rated power requires higher machine maintenance cost.

Table 8. R&M comparison of a 60 hp and 68 hp Combine Harvesters

Rated Power	Machine Cost, Php	% R&M/100 hr	Estimated R&M Cost, Php/hr	No of Operation, hr/year	Estimated R&M Cost, Php/year
59.1 hp	1,700,000.00	1.33%	226.1	480	108,528.00
68 hp	1,900,000.00	1.33%	252.7	480	121,296.00

Summary of Results/Recommendations:

1. An ideal test should be conducted with the same soil condition, crop condition, skills of operator, size and terrain of field, other technical parameters and environmental conditions. With these, the true fuel consumption and capacity can be determined. to establish the true field capacity and fuel consumption
2. Considering an operating speed of 4kph, 5kph and 6 kph and the width of cut ranges from 1.975m to 2.075m, the combine harvesters with power rating ranges from 60-88 hp have comparable theoretical field capacity in ha/hr.
3. Theoretical fuel consumption increases as the rated power of the combine harvester increases.
4. Higher machine rated power requires higher machine maintenance cost.
5. The theoretical fuel consumption and field capacity is a good reference of machine performance.

Prepared By:

MAYVILLE B. CASTRO
Technical Staff, FMFOD

RAYMOND R. ANDRES
Technical Staff, FMFOD

ALLEN DENVER V. MANGAOIL
Technical Staff, FMFOD

NIÑO D. BENGOSTA
Technical Staff, FMFOD

Reviewed By:

ALDRIN E. BADUA
Head, FMFOD

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